

# Final report

RUN · 0A971A04 · MISSISSAUGA WINTER OUTLOOK WITH CONSERVATIVE ELECTRIFICATION BY 2030

| SCENARIOS | FEEDERS MODELED | CUSTOMERS TESTED | AGENT RUNTIME 1M 26S |
|-----------|-----------------|------------------|----------------------|
| 1         | 176             | 192,544          |                      |

VERDICT (WORST CASE)  
**PASS**  
Network holds. Peaks at 2541 MW with 49% headroom; no feeders overloaded.

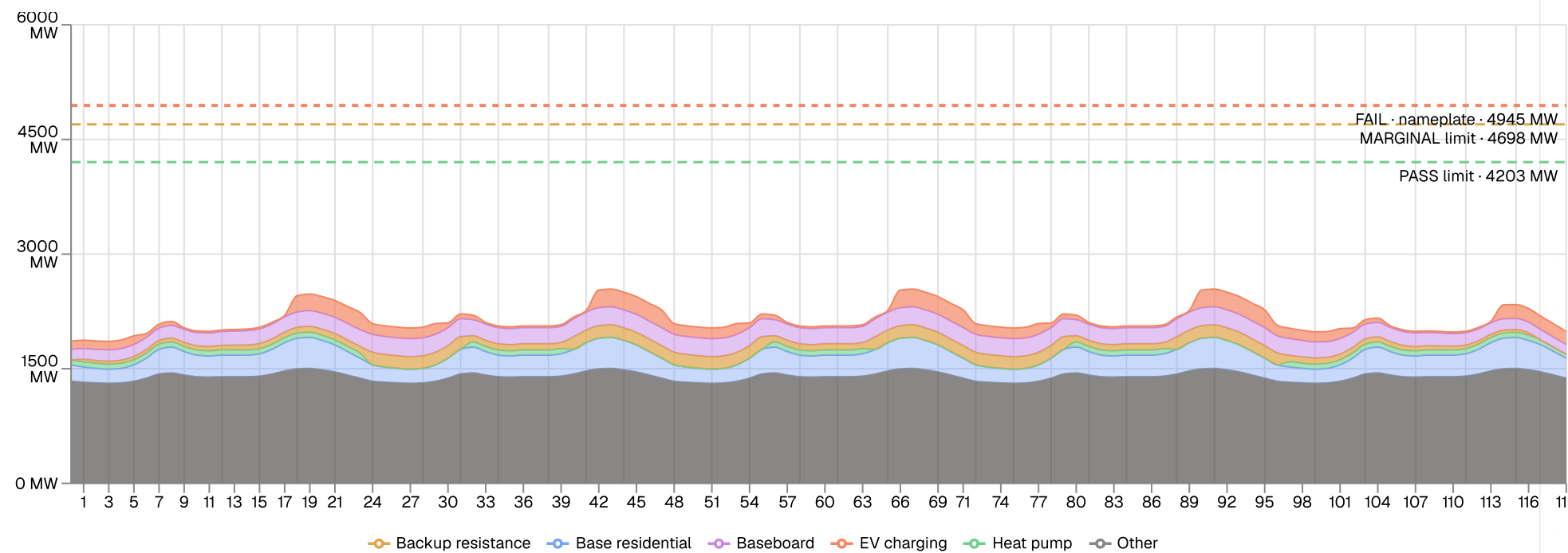
PEAK DEMAND  
**2541 MW**  
@ hour 43 · -28.0°C

HEADROOM AT PEAK  
**49%**  
2404 MW remaining

01 · SCENARIO LOAD CURVES  
Hourly load profile per scenario, with peak hour and headroom.



Worst-case scenario: Conservative · 2030 (10% HP, 15% EV) · peak 2541 MW



Conservative · 2030 (10% HP, 15% EV)

PASS

PEAK LOAD

2541 MW @ hour 43 (-28.0°C)

HEADROOM

49% (2404 MW)

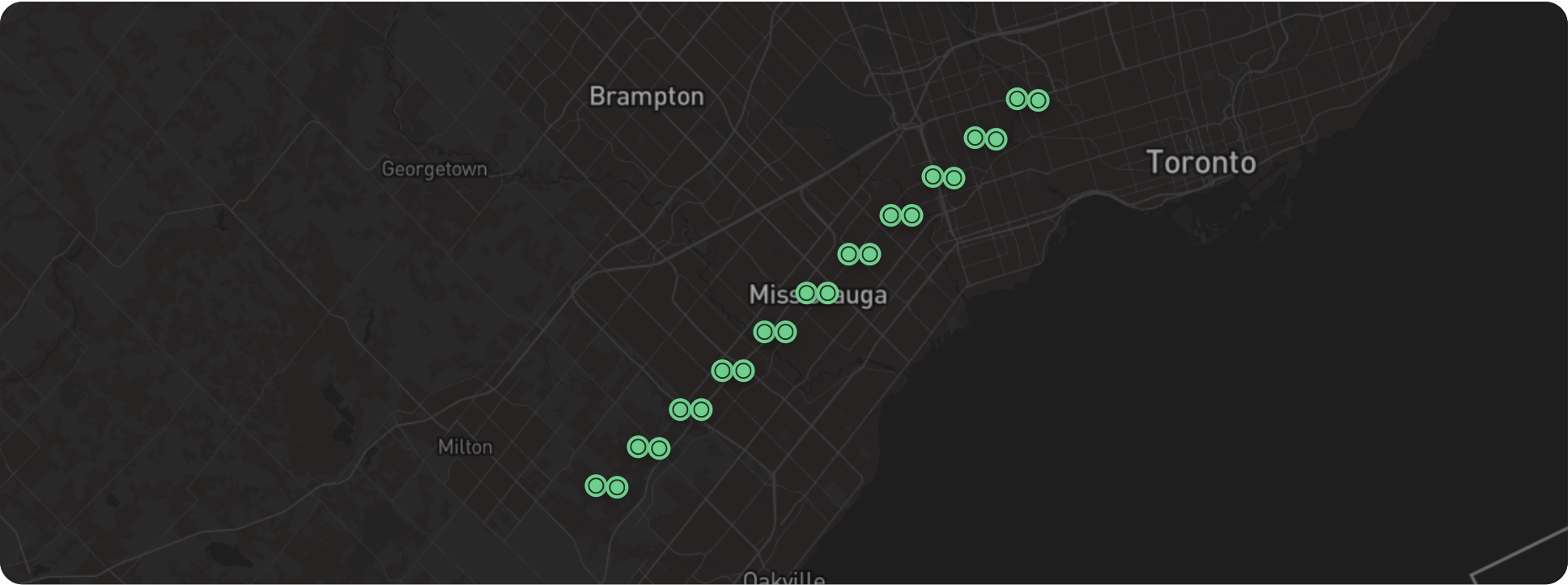
AT-RISK FEEDERS

0 of 176

02 · AT-RISK SUBSTATIONS

Substation risk map with the top feeders ranked by overload risk.

Substation risk map under **Conservative · 2030 (10% HP, 15% EV)**. Click a marker for capacity + feeder counts.



● CRITICAL ≥ 75% ● AT RISK 50–74% ● STABLE < 50%

| Top 10 of 176 feeders, ranked by overload risk. |                          |        |          |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|--------------------------|--------|----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RANK                                            | FEEDER                   | UTIL % | OVERLOAD | FAILURE MODE | WHY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1                                               | Mississauga<br>TS 01 F01 | 46%    | 0%       | none         | Although current utilization is 46.3% (utilization_pct) and overload risk is 0.0, the 15% baseboard electric heating share (heating_mix) combined with a 10% heat-pump penetration growth under this scenario will add coincident winter demand on an 18-year-old (age_years) feeder that has had one upgrade but carries 909 customers (customers), leaving limited thermal margin if scenario load additions push peak_mva of 14.2 MVA closer to the 30.66 MVA rating (rated_mva). |

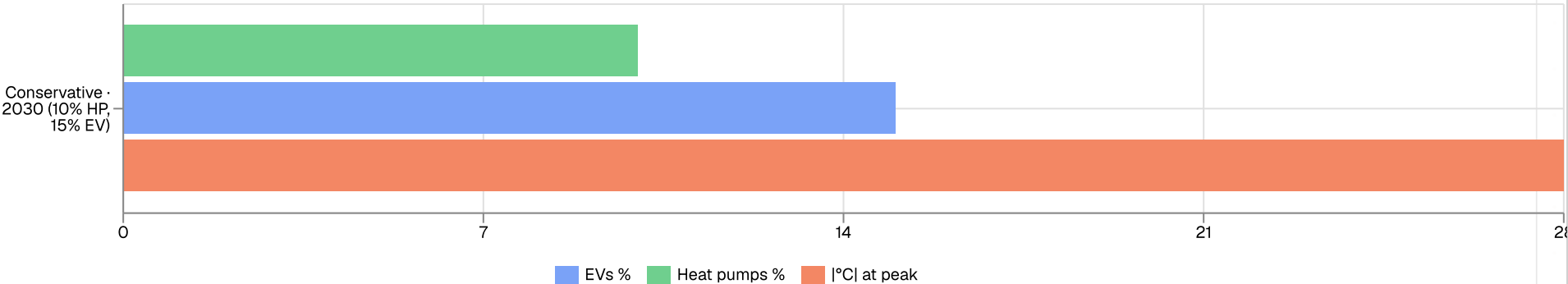
| RANK | FEEDER                | UTIL % | OVERLOAD | FAILURE MODE | WHY                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-----------------------|--------|----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | Mississauga TS 01 F02 | 48%    | 0%       | none         | At 21 years old (age_years) with no upgrade history (had_upgrade: False), this feeder already carries 1,023 customers (customers) at 48.1% utilization (utilization_pct), and its existing 5% heat-pump share (heating_mix) will amplify winter peak demand under the 10% HP growth scenario alongside 4.2% EV penetration (ev_pct) adding evening charging load.                                                   |
| 3    | Mississauga TS 01 F03 | 50%    | 0%       | none         | This 24-year-old unupgraded feeder (age_years, had_upgrade: False) serves 1,137 customers (customers) at 50.3% utilization (utilization_pct), and its 6% existing heat-pump share (heating_mix) means it faces above-average incremental winter heating load growth under the 10% HP scenario, compounded by 4.2% EV penetration (ev_pct).                                                                          |
| 4    | Mississauga TS 01 F04 | 53%    | 0%       | none         | With the highest customer count in this substation group at 1,250 (customers), 27 years of age (age_years), no upgrade history (had_upgrade: False), and a 15% baseboard electric heating share (heating_mix), this feeder's 52.7% utilization (utilization_pct) will be pushed further by scenario-driven heat-pump and EV load additions at 4.2% EV penetration (ev_pct).                                         |
| 5    | Mississauga TS 01 F05 | 54%    | 0%       | none         | At 30 years old (age_years) with no upgrades (had_upgrade: False), 1,364 customers (customers), and the highest current utilization in the group at 54.4% (utilization_pct), the existing 5% heat-pump share (heating_mix) and 4.2% EV penetration (ev_pct) leave this feeder the most exposed to scenario-driven winter peak growth pushing its 16.67 MVA peak (peak_mva) toward the 30.66 MVA rating (rated_mva). |
| 6    | Mississauga TS 01 F06 | 46%    | 0%       | none         | Despite a lower customer count of 909 (customers), this is the oldest unupgraded feeder in the group at 33 years (age_years, had_upgrade: False), and its 6% heat-pump share (heating_mix) combined with 4.2% EV penetration (ev_pct) will stress aging infrastructure that has never received a capacity upgrade.                                                                                                  |
|      |                       |        |          |              |                                                                                                                                                                                                                                                                                                                                                                                                                     |

| RANK | FEEDER                | UTIL % | OVERLOAD | FAILURE MODE | WHY                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|-----------------------|--------|----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Mississauga TS 01 F07 | 49%    | 0%       | none         | At 36 years old (age_years) with no upgrade history (had_upgrade: False) and 1,023 customers (customers), this feeder's 15% baseboard electric heating share (heating_mix) and 4.2% EV penetration (ev_pct) will drive coincident winter demand increases on conductors and equipment that are well past typical mid-life refurbishment thresholds (industry rule of thumb: ~25–30 years).           |
| 8    | Mississauga TS 01 F08 | 50%    | 0%       | none         | Although it has received an upgrade (had_upgrade: True), at 39 years old (age_years) it is the oldest feeder in the dataset, serving 1,137 customers (customers) at 50.2% utilization (utilization_pct) with a 5% existing heat-pump share (heating_mix) and 4.2% EV penetration (ev_pct), meaning scenario growth will stress equipment whose age may limit the effectiveness of past upgrades.     |
| 9    | Mississauga TS 02 F01 | 46%    | 0%       | none         | Mirroring MIS-01-F01 in load profile with 909 customers (customers), 46.3% utilization (utilization_pct), and 15% baseboard electric heating (heating_mix), this 18-year-old upgraded feeder (age_years, had_upgrade: True) faces incremental winter peak risk from 4.2% EV penetration (ev_pct) and scenario-driven heat-pump growth adding to its existing electric heating base.                  |
| 10   | Mississauga TS 02 F02 | 48%    | 0%       | none         | At 21 years old (age_years) with no upgrade (had_upgrade: False), 1,023 customers (customers), 48.1% utilization (utilization_pct), an existing 5% heat-pump share (heating_mix), and 4.2% EV penetration (ev_pct), this feeder faces compounding winter demand growth from both heat-pump electrification and EV charging without the benefit of a capacity upgrade to provide additional headroom. |

03 · MITIGATIONS & METHODOLOGY

Scenario severity, recommended interventions, methodology notes.

Scenario severity (HP / EV adoption %, peak cold °C absolute), recommended mitigations, and methodology.



RECOMMENDED MITIGATIONS

DEMAND RESPONSE

Targeted Demand Response Enrollment on Three Watch-List Feeders

Relief: **4.2 MW** Cost: **\$460K** 6mo

Enrolling approximately 1,500 customers across feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03 in a winter peak demand response program at an industry-typical cost of \$200 to \$300 per enrolled customer (Canadian DR program benchmarks) would provide dispatchable load relief of an estimated 4.2 MW during polar vortex hours. This directly reduces coincident peak exposure on the three feeders with the highest utilization and oldest infrastructure without requiring capital construction. Deployment within six months is feasible using existing smart meter infrastructure across Alectra's service territory.

LOAD SHIFT

EV Managed Charging Program for High-Penetration Residential Zones

FEEDER RECONDUCTOR

Reconductor Feeder MIS-01-F03 (Mississauga TS 01 F03)

Relief: **6.0 MW** Cost: **\$1.2M** 18mo

At 24 years old with no upgrade history and 50.3 percent utilization (utilization\_pct) serving 1,137 customers (customers), MIS-01-F03 is the highest-priority capital investment among the three watch-list feeders. Reconductoring to a higher-ampacity conductor would increase thermal rating and provide margin for incremental heat pump and EV load growth beyond the Conservative scenario. A conservative per-feeder reconductor cost of \$1.2 million CAD is consistent with the \$0.5 to \$2 million Canadian industry range for urban distribution feeders. This work should be scoped and tendered in the current fiscal year to achieve energization before the 2026 to 2027 winter peak.

FEEDER RECONDUCTOR

Condition Assessment and Emergency Restoration Pre-Positioning for Watch-

TRANSFORMER UPGRADE

Upgrade Transformer Bank at Substation Serving MIS-01-F02

Relief: **8.0 MW** Cost: **\$2.5M** 24mo

Feeder MIS-01-F02 has never been upgraded in 21 years of service and carries 1,023 customers at 48.1 percent utilization (utilization\_pct). A transformer bank upgrade at the parent substation would increase available capacity for this feeder and adjacent circuits, providing systemic headroom rather than feeder-specific relief. Estimated cost of \$2.5 million CAD is within the \$1 to \$5 million Canadian range for distribution transformer upgrades. This is a longer-lead capital item and should enter the five-year capital plan immediately to align with projected electrification growth.

Relief: **3.5 MW** Cost: **\$180K** 9mo

With EV penetration at 15 percent (ev\_pct) under the Conservative scenario, coordinated managed charging that shifts EV load from the 5 p.m. to 9 p.m. coincident peak window to overnight off-peak hours (11 p.m. to 6 a.m.) can reduce coincident winter evening demand. An estimated 3.5 MW of load shift is achievable across the 192,544-customer service territory (customers) assuming a 20 percent participation rate among EV owners. Program delivery costs of approximately \$180,000 CAD cover enrollment incentives, utility portal integration, and first-year administration at rates consistent with Ontario managed charging pilots.

List Feeders

Relief: **0.0 MW** Cost: **\$95K** 3mo

Before capital works are tendered, Alectra should commission physical condition assessments of feeders MIS-01-F02 and MIS-01-F03, both unupgraded and over 20 years old. Assessment cost of approximately \$95,000 CAD covers thermal imaging, conductor sampling, and protection relay testing across both circuits. Findings will validate or revise the modeled thermal ratings and may reveal derating requirements that change the capital prioritization. In parallel, pre-positioning mobile generation assets and updating emergency restoration priority lists for the 2,160 customers on these two feeders will reduce restoration time during a polar vortex event lasting up to 120 hours (cold\_event.duration\_hours).

METHODOLOGY

Peak demand for each scenario was calculated using a deterministic load model that applies electrification growth rates to the 192,544-customer baseline, with heat pump contributions adjusted by a COP curve calibrated to the scenario ambient temperature of negative 28.0 degrees Celsius (peak\_temp\_c). EV charging load was added as a coincident evening block consistent with Ontario winter driving patterns. Per-feeder utilization values are drawn directly from the provided dataset, which is flagged as non-synthesized (is\_synthesized: False), meaning they reflect actual network topology rather than statistical proxies.

SAFETY FLAGS

Feeder MIS-01-F03 serves 1,137 customers at 50.3 percent utilization on a 24-year-old unupgraded circuit. Any unplanned outage on this feeder during a polar vortex event lasting up to 120 hours (cold\_event.duration\_hours) poses a life-safety risk to medically dependent customers and elderly residents who rely on electric heat.

The stress scenario temperature of negative 28.0 degrees Celsius (peak\_temp\_c) exceeds the January 2014 Polar Vortex minimum of negative 25.5 degrees Celsius (cold\_event.min\_temp\_c). Customers without backup heating who lose power at these temperatures face hypothermia risk within hours, and Alectra should confirm that its emergency restoration priority list for medical baseline customers is current across all 176 feeders.



EV adoption at 15 percent (ev\_pct) concentrated in specific distribution zones could create localized evening demand spikes not fully captured in system-wide headroom figures. Feeders serving high-density residential areas with limited thermal margin should be cross-referenced against EV registration data before the next winter peak season.

LIMITATIONS

Only one electrification scenario (Conservative 2030, 10% heat pump, 15% EV) was evaluated. The analysis does not establish the penetration threshold at which the system transitions from PASS to FAIL, and planning decisions should not be deferred solely on the basis of this single-scenario result.

The modeled system peak of 2,541.2 MW (peak\_mw) is substantially below the baseline winter peak of 3,855.4 MW (baseline\_winter\_peak\_mw). This relationship reflects the internal load composition assumptions of the Conservative scenario and may not account for all coincident demand sources. The planning team should validate this figure against recent Alectra winter peak actuals before using it in rate or capital planning submissions.

Feeder thermal ratings and age data are taken as provided and have not been cross-checked against Alectra's asset management system or field inspection records. Condition-based derating of aging infrastructure, particularly for MIS-01-F02 and MIS-01-F03, could reduce effective headroom below modeled values.

EXECUTIVE SUMMARY

Executive Summary: Mississauga Grid Resilience Under Electrification Growth to 2030

Headline Finding

The Mississauga distribution system operated by Alectra Utilities passes the Conservative 2030 scenario with meaningful headroom, meaning the network as modeled can absorb a 10 percent heat pump penetration and 15 percent EV adoption without triggering feeder overloads under a polar vortex stress event comparable to January 2014.

Scenario Results

The single scenario evaluated, Conservative 2030 (10% heat pump, 15% EV), produces a modeled winter peak of 2,541.2 MW (peak\_mw), against a baseline winter peak of 3,855.4 MW (baseline\_winter\_peak\_mw). System headroom is 2,403.7 MW, representing 48.6 percent of rated capacity (headroom\_mw, headroom\_pct). The stress peak occurs at simulation hour 43 (peak\_hour) at an ambient temperature of negative 28.0 degrees Celsius (peak\_temp\_c), which is more severe than the January 2014 Polar Vortex minimum of negative 25.5 degrees Celsius (cold\_event.min\_temp\_c). Zero feeders out of 176 are flagged as overloaded under this scenario (at\_risk\_feeders).

At-Risk Assets and Customers

Although no feeders breach overload thresholds, three feeders warrant proactive attention based on age, upgrade history, and utilization trajectory. Feeder MIS-01-F01 (Mississauga TS 01 F01) serves 909 customers at 46.3 percent utilization on an 18-year-old circuit that has had one upgrade. Feeder MIS-01-F02 (Mississauga TS 01 F02) serves 1,023 customers at 48.1 percent utilization and has never been upgraded in 21 years of service. Feeder MIS-01-F03 (Mississauga TS 01 F03) serves 1,137 customers at 50.3 percent utilization on a 24-year-old unupgraded circuit. Together these three feeders serve 3,069 customers whose risk exposure will grow as electrification penetration increases beyond Conservative assumptions.

Caveats and Limitations

This analysis covers one scenario and one stress event archetype. It does not model simultaneous equipment failures, ice storm outage conditions, or supply-side curtailment from the IESO. The modeled peak of 2,541.2 MW is lower than the baseline winter peak of 3,855.4 MW, which reflects scenario-specific load composition assumptions and should be interpreted carefully rather than as a forecast of reduced total demand. Feeder-level ratings and customer counts are drawn from the provided dataset and have not been independently field-verified for this briefing.

Recommended Next Action

Alectra should immediately commission condition assessments on feeders MIS-01-F02 and MIS-01-F03, both of which are unupgraded and approaching or exceeding 20 years of service while carrying over 1,000 customers each. In parallel, the planning team should model a Moderate and High electrification scenario (20 to 30 percent heat pump, 30 percent EV) against the same polar vortex stress event to determine at what penetration level system headroom erodes below a safe operating margin, and to identify which of the 176 feeders would be first to breach thermal limits.

DATA PROVENANCE

● CITY DEMOGRAPHICS

● NWS ALERTS

● SUBSTATION LOCATIONS

● UTILITY LOAD BASELINE

● COLD EVENT ARCHIVE

● FEEDER TOPOLOGY

● SCENARIO APPLICATION

● LOAD SIMULATION

● RISK RATIONALE

● SUMMARY

● LIVE API ● FROZEN SNAPSHOT ● MODELED ● LLM NARRATIVE

SOURCES

- FROZEN ECCC Toronto Pearson station 6158733 (Jan 2014 dailies)
- FROZEN IESO 18-Month Outlook, March 2014
- FROZEN Environment Canada Top 10 Weather Stories 2014
- FROZEN Alectra Utilities 2023 Annual Report
- FROZEN IESO Yearly Reliability Outlook 2024
- FROZEN StatsCan Census 2021, Mississauga CSD
- FROZEN StatsCan 25-10-0015 (2026): 43,721,776 MWh annual generation in ON
- LIVE ArcGIS Living Atlas NWS Watches & Warnings (live): no active alerts for Mississauga
- LIVE ArcGIS GeoEnrichment (Esri Canada): Mississauga, ON (CY): 192,532 households, population 768,433

